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COMPARATIVE SOCIAL MEDIA USE AND THE MENTAL HEALTH OF MOTHERS WITH HIGH LEVELS OF PERFECTIONISM

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Social media is thought to communicate idealized images and discourses of motherhood. As such, it may present as a risk factor for poor mental health in mothers who strive for perfection and compare themselves to the ideals presented on social media. The present study examined the influence of Facebook and Instagram on the relationship between perfectionism in mothers and their mental health. A sample of 201 mothers completed an online survey. Two dimensions of perfectionism were assessed: Self-Orientated Parenting Perfectionism (SOPP) and Societal-Prescribed Parenting Perfectionism (SPPP). Mediation models were conducted to examine social media frequency and social comparison respectively on the relationship between perfectionism and maternal mental health. Results revealed that for mothers with SOPP, the process of social comparison with other mothers on social media contributed to symptoms of anxiety and depression. The amount of time engaging in social media however, had no impact. In contrast, for mothers with SPPP, the amount of time spent on Facebook contributed to symptoms of depression and anxiety, while the process of social comparison led to anxiety symptoms alone. Social comparison appears to be important for perfectionistic mothers who use social media, as this may contribute to negative mental health outcomes.

Keywords: perfectionism, depression, anxiety, stress, parenting, social media

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The transition to parenthood is argued to be one of the most stressful life events and represents a time when new mothers are at increased risk of developing mood and anxiety disorders (Milgrom et al., 2008; Milgrom, Westley, & Gemmill, 2004). Evidence has linked these disorders with detrimental outcomes for the mother and her baby, such as lower-self efficacy in mothers and impaired cognitive, social, and emotional development in children (Milgrom et al., 2004; Murray & Cooper, 1997). Risk factors for the development of maternal depression and anxiety include genetic factors, personal and family psychiatric history, prenatal anxiety and depression, poor social support, and neuroticism (Milgrom et al., 2004; Robertson, Grace, Wallington, & Stewart, 2004). Identifying the risk factors for poor maternal mental health is important as it may help to guide future treatment and reduce the impact of depression and anxiety on the mother and child (Milgrom et al., 2004; Seymour, Giallo, Cooklin, & Dunning, 2014).

Maternal wellbeing has also been linked with perfectionism. Specifically, mothers with perfectionistic traits have been shown to display poorer parental adjustment, increased anxiety, and lowered self-efficacy (Gelabert et al., 2012; Lee, Schoppe-Sullivan, & Kamp Dush, 2012). According to Hewitt and Flett (1991) perfectionism is a multidimensional construct comprised of three dimensions; self-orientated, other-orientated and socially prescribed perfectionism. Snell, Overbey, and Brewer (2005) expanded these dimensions to parenting and developed self-oriented parenting perfectionism and societal-prescribed parenting perfectionism (Snell et al., 2005). Self-oriented parenting perfection reflects the degree to which one holds excessively high standards for one's own parenting, while societal-prescribed parenting perfectionism involves the belief that society holds high standards for their parenting (Snell et al., 2005). Lee et al. (2012) found that societal-prescribed parenting perfectionism was associated with lower levels of self-efficacy, while self-orientated parenting perfectionism was associated with an increase in parenting satisfaction, suggesting that self-orientated perfectionism may be adaptive in mothers. However, perfectionism was only assessed during the third trimester of pregnancy, so there remains a need to investigate perfectionism following childbirth.

THE ROLE OF MASS MEDIA

The pressure to be a perfect mother is thought to be shaped by the mass media, which communicates idealized images and discourses of motherhood (Gentile, 2011; Henderson, Harmon, & Newman, 2016). These discourses may be problematic for parents who use them as a benchmark for comparison (Choi, Henshaw, Baker, & Tree, 2005; Festinger, 1954). The recent growth of easily accessible social networking sites has further facilitated social comparison behaviors (Pempek, Yermolayeva, & Calvert, 2009). Given the link between perfectionism and poor maternal mental health, social networking sites such as Facebook and Instagram may present as a particular risk factor for mothers who strive for perfection and compare themselves to the ideals presented on social media.

SOCIAL MEDIA AND SOCIAL COMPARISON

Increased accessibility to the internet and online parenting sites has provided mothers with another medium to gain and share information on motherhood and parenting. For instance, Plantin and Daneback (2009) found that new mothers (aged 30–35) were the most active internet users when looking up health and parenting information. It was also found that mothers who identified as lacking support from their own parents, friends, and relatives, access support though online communities (Hall & Irvine, 2009).

However, while some sites provide support and information sources for parents, other sites may reinforce traditional gender stereotypes and unequal gender roles, which may contribute to parenting stress in mothers (Madge & O'Connor, 2006). Social networking sites may be especially problematic, particularly when individuals spend most of their time viewing idealized profiles and pictures of others (Pempek et al., 2009). Typically, users on these sites promote the most positive representations of themselves and their lives (Dorethy, Fiebert, & Warren, 2014;

Gonzales & Hancock, 2011), perhaps partly to avoid being perceived as a bad mother (Choi et al., 2005). Therefore, they become an easily accessible platform for social comparison, where individuals can compare themselves with others on characteristics such as appearance, success, and parenting ideals (Lee, 2014; Lee et al., 2012).

According to social comparison theory, individuals are likely to make positive or negative self-evaluations by comparing themselves with others (Festinger, 1954; Gibbons & Buunk, 1999) and are more likely to do this in times of stress, novelty, or change. Lee (2014) found that college students used Facebook to view other profiles and photos and make comparisons. Students with increased uncertainty about themselves and reduced self-esteem appeared to engage in greater social comparison and experienced more negative feelings. The impact of social comparison on mental health symptoms was not investigated however. Given that parenting is one of the most stressful life events with greater uncertainty and significant change (Milgrom et al., 2008), mothers may be at particular risk of comparing themselves against ideals presented on social media. Therefore, comparing oneself with images of the perfect mothering ideal may be problematic for the mental health of mothers, particularly those who strive to attain perfection.

FACEBOOK

The literature regarding the relationship between the frequency and duration of Facebook use and mental health remains inconclusive. Increased amounts of time spent on Facebook may result in negative emotions and stress (Bartholomew, Schoppe-Sullivan, Glassman, Kamp Dush, & Sullivan, 2012). However, other studies have not found a relationship between duration of online social networking use—including Facebook—and depressive symptoms (Jelenchick, Eickhoff, & Moreno, 2013).

Instead of frequency and duration of Facebook use, it may in fact be the process of social comparison which puts a person at risk of mental health difficulties. For instance, some studies have revealed an increase in depression symptoms in undergraduate

students, after spending increased time on Facebook as they engaged in social comparison (Steers, Wickham, & Acitelli, 2014). Thus, not only the frequency of Facebook use, but also engagement in social comparison may influence mental health.

Although limited in number, studies have emerged examining new mothers' engagement with Facebook use. For instance, a recent study found that new mothers engaged in Facebook to promote images and behaviors that align with the motherhood ideal (Schoppe-Sullivan et al., 2017). More specifically, Schoppe-Sullivan et al. (2016) assessed societal-prescribed parenting perfectionism and Facebook use in mothers. According to their findings, it was not the quantity of time spent on Facebook, rather the quality of the interaction that impacted on the mental health of new mothers. Their sample reported feeling pressure to "meet society's impossibly high standards for parenting" (p. 286) by increasing their Facebook activity, such as posting photos of their baby and mothering practices, to receive confirmation for meeting societal ideals of motherhood. As such, they experienced positive emotions when their mothering was validated and negative emotions when validation was not offered. Thus, the researchers concluded that the emotional experiences of new mothers were contingent on external validation.

INSTAGRAM

While Facebook is still the most popular social networking site, Instagram, created in 2010, is reported to be the fastest growing social networking site. Instagram is distinct from Facebook in that it is a platform that allows very little text, thereby focusing on sharing images, creating a strong visual-orientated culture (Lup, Trub, & Rosenthal, 2015), where only the best aspects of one's self and life are presented (Marwick, 2015).

According to Marwick (2015), the most followed users on Instagram are celebrities, and images of celebrity mothers "are perfect, most of them white, always rich, happy, hands-on, and in control" (p. 1, Douglas & Michaels, 2000, 2004). A recent qualitative study found that mothers felt pressured to match the motherhood ideal on Instagram, therefore they "portrayed unrealistic

and at times idealistic images of the family and of themselves" (p. 24, Djafarova & Trofimenko, 2017), and rarely discussed the daily challenges of parenting. Furthermore, Lup et al. (2015) discovered that individuals who followed more strangers on Instagram, engaged in social comparisons, and therefore were at greater risk of experiencing depressive symptoms and negative self-judgments. Thus, there is emerging evidence to suggest that mothers compare themselves to others on Instagram and this has a negative impact on their emotional wellbeing.

MATERNAL MENTAL HEALTH AND SOCIAL SUPPORT

Together, perfectionism and engagement in social comparison may be particularly pernicious for the mental health of mothers. This is especially problematic given that maternal depression can impact negatively on the interaction between mother and child and impair the cognitive, emotional, and social development of the child (Milgrom et al., 2004; Murray & Cooper, 1997). Similarly, research on postnatal anxiety has shown that high levels of maternal anxiety can have adverse effects on parent-child interactions and on the development and behaviors of the child (Seymour et al., 2014). Thus, if maternal mental health is undermined by perfectionism and social comparison, then this is important for not just the mother themselves, but also for child attachment and their overall development.

Social support, defined as "an interpersonal transaction that provides individuals with esteem, stress-related aid, and emotional assistance" (p. 278, Haslam, Pakenham, & Smith, 2006), has been found to be a significant protective factor against the development of mood and anxiety disorders in mothers. Higher levels of partner support and extended social support following birth appear to reduce stress and anxiety and may protect mothers from developing depressive symptoms (Collins, Dunkel-Schetter, Lobel, & Scrimshaw, 1993; Haslam et al., 2006; Levy-Shiff, Dimitrovsky, Shulman, & Har-Even, 1998). In this respect, levels of social support may serve to moderate the relationship between social media use and depressive symptoms in mothers.

THE PRESENT STUDY

To the best of our knowledge, the present study is the first to investigate whether Facebook and Instagram use mediates a well-established link between perfectionism and mental health in mothers. In this paper, mental health is considered to refer to symptoms of depression, anxiety, and stress, although we acknowledge that these do not capture the full scope of mental health as it is typically defined. This is particularly important as poor maternal mental health is linked to unfavorable outcomes for the mother and her baby (Milgrom et al., 2004; Murray & Cooper, 1997; Seymour et al., 2014). It was hypothesized that:

1. Perfectionism in mothers (both SOPP and SPPP) would be positively associated with depression, anxiety, and stress symptoms, respectively.
2. After controlling for social support, the relationship between perfectionism in mothers (both SOPP and SPPP) and depression, anxiety, and stress, respectively, would be weaker. In other words, social support may in part explain some of the variance in depression, anxiety, and stress symptom scores respectively, such that accounting for social support would reduce the extent of covariation between perfectionism and each of these symptom domains.
3. Facebook and Instagram frequency would not mediate the relationship between perfectionism (SOPP and SPPP) in mothers and maternal mental health, after partialling out the influence of social support, consistent with the possibility that it may in fact be the process of social comparison, rather than frequency of use, which has pernicious effects on mental health.
4. The relationship between perfectionism (SOPP and SPPP) in mothers and maternal mental health would be mediated by social comparison when engaging in Facebook and Instagram, after partialling out the influence of social support.

METHOD

PARTICIPANTS

Mothers aged between 22 and 53 years (N = 250; M = 37.38, SD = 5.04) were recruited for the study.

SELF-REPORT QUESTIONNAIRES

The Depression Anxiety Stress Scale-21 (DASS-21). The DASS-21 (Lovibond & Lovibond, 1995a, 1995b) consists of 21 items that assess the negative emotional states of depression, anxiety, and stress over the last week. Items are rated on a four-point Likert-type scale ranging from 1(never) to 4 (almost always). High scores indicate greater levels of anxiety, stress, and depressive symptoms. According to Henry and Crawford (2005), the DASS-21 revealed good to excellent reliability and internal consistency for all three factors (depression = .88, anxiety = .82, and stress = .93). This study also revealed good to excellent Cronbach's alpha, for depression = .91, anxiety = .81, and stress = .86

The Iowa-Netherlands Comparison Orientation Measure (INCOM). The INCOM (Gibbsons & Buunk, 1999) is an 11-item measure assessing individual differences when comparing themselves to others. Chae (2015) adapted the items to include the word mother, to assess their degree of social comparison. The present study expanded on this to assess the level of social comparison mothers engage in when comparing themselves to "other mothers/families on Instagram and/or Facebook." A sample item includes: "When I look at Instagram and/or Facebook, I often compare myself with other mothers/families with respect to what I have accomplished as a parent." Participants rated their level of agreement with each statement on a five-point Likert-type scale ranging from 1 (I strongly disagree) to 5 (I strongly agree). High score indicated increased levels of social comparison. According to Gibbsons and Buunk (1999), the INCOM has good psychometric properties, demonstrating good internal consistency (= .83). The internal consistency for this study was good (= .88).

The Multidimensional Parenting Perfectionism Questionnaire (MPPQ). The MPPQ (Snell et al., 2005) is a 65-item questionnaire that consists of 11 subscales that measure different aspects of parenting perfection. This study included two of the subscales (12-items), Societal-Prescribed Parenting Perfectionism (SPPP) and Self-Orientated Parenting Perfectionism (SOPP). SPPP measures the degree in which one believes society expects them to be a perfect parent, whereas, SOPP assesses the high standards one places on oneself to be a perfect parent. Items are rated using a five-point Likert-type scale ranging from 1 (not at all characteristic of me) to 5 (very characteristic of me). Higher scores indicate high parenting perfection for the self (SOPP) and that society holds high standards for their parenting (SPPP). A sample item for SPPP includes: "Most people in society expect me to always be a perfect parent." A sample item for SOPP includes: "I set very high standards for myself as a parent." Both SPPP and SOPP scales have shown good to excellent internal consistency in previous studies, ranging from .84 to .91 (Snell et al., 2005). In this study, the internal consistency for SOPP was excellent (= .91) and good (= .88) for SPPP.

The Multidimensional Scale of Perceived Social Support (MSPSS). The MSPSS (Zimet, Dahlem, Zimet, & Farley, 1988) is a 12-item measure assessing perceived social support from three sources: Family, Friends, and Significant Other. Participants rated their level of agreement on a seven-point Likert-type scale ranging from 1 (very strongly disagree) to 7 (very strongly agree). Higher scores indicate increased levels of perceived social support from Family, Friends, and Significant Other. According to Zimet, Powell, Farley, Werkman, and Berkoff (1990), the MSPSS is a psychometrically sound measure, revealing good to excellent internal reliability (= .84 to .94). The Cronbach's alpha for this measure in this study was excellent (= .94).

Social Media Exposure items: To determine the extent to which the participants frequented the social media sites Facebook and Instagram, they were asked to rate on average how many hours per day they spent observing and/or engaging in each of these medias using a seven-point Likert-type scale, ranging from 1 (less than 1 hour per day) to 7 (6+ hours per day).

PROCEDURE

Participants were recruited via the social networking site, Facebook, where regular posts were uploaded to the Facebook pages of mother groups (Sydney "North Shore Mums" and Sydney "Northern Beaches Mums"), and the Facebook page of the Gidget Foundation, an organization that targets perinatal and postnatal anxiety and depression. Each post included the link to the online survey, which was administered using the Qualtrics web-based system. Approval for the study was obtained from the University of Technology Sydney Human Research Ethics Committee (HREC Ref no. 2015000482-52) and all participants provided informed consent.

DATA ANALYSIS

The Statistical Package for Social Sciences (SPSS) version 24 was used for all analyses. The distributions of variables were inspected and descriptive statistics were calculated. Multiple mediation models were used to examine whether the relationship between perfectionism and maternal mental health (depression, anxiety, stress) was mediated by the frequency of social media use (Facebook and Instagram) and social comparison, respectively after controlling for social support.

RESULTS

SAMPLE

Of the 250 survey respondents, 49 did not complete the survey, and were therefore excluded from the study. The final sample consisted of 201 adult mothers. Sixty participants (29.7%) were first time mothers, 93 (46.0%) were second time mothers, with the remainder having had three or more children (n = 40, 19.8% had 3 children; n = 4, 6% had 4 children; n = 2, 1.0% had more than 4 children). In terms of relationship status, 157 (77.7%) mothers reported that they were married, 26 (12.9%) reported that they were in a de facto relationship, 9 (4.5%) reported that they were single and 9 (4.5%) reported other. A majority of mothers had

TABLE 1. Summary Of Pearson Correlations For Key Variables

	Measures	1	2	3	4	5	6	7	8	9
1.	Depression	—								
2.	Anxiety	.53**	—							
3.	Stress	.65***	.66***	—						
4.	FB Frequency	.23**	.27***	.16*	—					
5.	Instagram Frequency	.09	.06	.00	37***	—				
6.	SOPP	.22**	.30***	.33***	.12	.09	—			
7.	SPPP	.35***	.32***	.38***	.18*	.14	.76***	—		
8.	Social Support	−.47***	−.28***	−.31***	−.11	.05	−.10	−.19**	—	
9.	Social Comparison	.22**	.24**	.20**	.36***	.15*	.32***	.42***	−.09	—

Note. No of children = Number of children; FB Frequency = Facebook Frequency; SOPP = Self-Orientated Parenting Perfectionism; SPPP = Societal-Prescribed Parenting Perfectionism.
*$p < .05$; **$p < .01$; ***$p < .001$.

completed an undergraduate university degree (105; 52.2%) and a further 52 (25.8%) had achieved a post-graduate university degree. Seventy-one mothers (35.1%) were working part-time and 66 (32.7%) employed full-time. The majority of the participants were Caucasian (N = 86, 92.1%) and 77 (38.1%) revealed having a mental health diagnosis in the past, of which 37 (18.3%) reported having a current mental health diagnosis. The average time the sample reported engaging in Facebook frequency equalled 1 hour per day, whereas, the average time spent frequenting Instagram per day, equalled 17.9 minutes.

Correlation Analysis

Pearson correlations were conducted to assess the relationships between maternal mental health (depression, anxiety, and stress) and each of the key variables: frequency of social media engagement (Facebook and Instagram), social comparison when using social media, perfectionism (SPPP and SOPP), and social support (see Table 1). All correlations were assessed using a two-tailed test of significance, $p < .05$ and Cronbach alpha levels did not suggest the presence of multicollinearity, as all values were < .80 (Field, 2013).

As can be seen in Table 1 and in line with the first hypothesis, both perfectionism variables (SOPP and SPPP) were correlated with DASS-21 scores (weak to moderate positive correlations). Additionally, social support was significantly negatively corre-

TABLE 2. Summary of Partial Correlations when controlling for Social Support

Measures	1	2	3	4	5	6	7	8
1. Depression	—							
2. Anxiety	.47***	—						
3. Stress	.60***	.63***	—					
4. FB Frequency	.21**	.25***	.13	—				
5. Instagram Frequency	.13	.08	.02	.38***	—			
6. SOPP	.20**	.28***	.31***	.11	.09	—		
7. SPPP	.30***	.28***	.35***	.16*	.15*	.76***	—	
8. Social Comparison	.20**	.23***	.19**	.35***	.15*	.32***	.41***	—

Note. No of children = Number of children; FB Frequency = Facebook Frequency; SOPP = Self-Orientated Parenting Perfectionism; SPPP = Societal-Prescribed Parenting Perfectionism.
*$p < .05$; **$p < .01$; ***$p < .001$.

lated with maternal mental health (weak to moderate correlations), suggesting that as social support increases, mothers report lower levels of depression, anxiety, and stress. This is also evident with SPPP, whereby mothers report less concern that society holds high standards for their parenting with increased social support.

Of the media exposure variables, Facebook frequency had a significant (weak, positive) correlation with Instagram exposure, SPPP, social comparison, and DASS-21 scores. This suggests that mothers who rate themselves as frequently engaging in Facebook, also engage in Instagram and compare themselves to other mothers and families when using social media. These mothers also endorsed higher levels of depression, anxiety, and stress symptoms.

As can be seen in Table 2, and as predicted, after controlling for social support, the relationships between each of SOPP and SPPP and DASS-21 scores remained significant and were weaker (weak to moderate correlations). Therefore, social support appears to influence the relationship between dimensions of perfectionism and depression, anxiety and stress, however to a small extent.

Further examination of the partial correlations table revealed that the relationship between Facebook frequency and DASS-21 stress was no longer significant after partialling out social support. Dissimilarly, SPPP was significantly correlated with Instagram frequency when partialling out the influence of social sup-

TABLE 3. Mediation Analysis for Facebook and Instagram Frequency, when Partialling Out Social Support

				Indirect effect		BCa 95% CI	
IV	**Mediator**	**DV**	R^2	***ab***	***SE B***	**Lower**	**Upper**
SOPP	Facebook	Depression	.25	.01	.01	−.00	.04
SOPP	Facebook	Anxiety	.15	.01	.01	−.00	.04
SOPP	Facebook	Stress	.18	.01	.01	−.00	.03
SPPP	Facebook	Depression	.29	.02	.01	.00	.04
SPPP	Facebook	Anxiety	.15	.02	.01	.00	.05
SPPP	Facebook	Stress	.20	.01	.01	−.00	.03
SOPP	Instagram	Depression	.25	.01	.01	−.00	.03
SOPP	Instagram	Anxiety	.15	.00	.00	−.00	.01
SOPP	Instagram	Stress	.18	-.00	.00	−.01	.01
SPPP	Instagram	Depression	.29	.01	.01	−.00	.03
SPPP	Instagram	Anxiety	.15	.00	.01	−.01	.02
SPPP	Instagram	Stress	.02	-.00	.01	−.02	.01

Note. CI = Confidence Interval; SOPP = Self-Orientated Parenting Perfectionism; SPPP = Societal-Prescribed Parenting Perfectionism.

port, although to a weak extent. Therefore, mothers with beliefs that society holds high expectations for their parenting are significantly influenced by Instagram frequency, after controlling for the influence of social support.

Mediation Analyses

Mediation analyses were conducted using the PROCESS macro in SPSS (Hayes, 2013) and based on a bootstrap approach using 10,000 (re)samples. Bootstrapping is suggested to have greater power for small samples (were $n < 400$; MacKinnon et al., 2013) and to be appropriate when the product of coefficients is not normally distributed. Indirect effects (*ab*) were considered significant when the bias-corrected and accelerated (BCa) 95% Confidence Intervals (CI) did not include zero (Hayes, 2013).

Hypothesis 3 predicted that the mental health of mothers with perfectionism would not be mediated by social media frequency. As hypothesized, the analyses showed that Facebook frequency did not mediate the relationship between mothers with SPPP and stress (*ab* = .01, BCa 95% CI [−.00, .03]), or between SOPP

TABLE 4. Mediation analysis for Social Comparison, When Partialling Out Social Support

				Indirect effect		BCa 95% CI	
IV	**Mediator**	**DV**	R^2	***ab***	***SE B***	**Lower**	**Upper**
SOPP	Social Comparison	Depression	.25	.03	.02	.00	.07
SOPP	Social Comparison	Anxiety	.15	.02	.01	.00	.05
SOPP	Social Comparison	Stress	.18	.02	.01	-.01	.05
SPPP	Social Comparison	Depression	.29	.02	.02	-.02	.07
SPPP	Social Comparison	Anxiety	.15	.03	.02	.00	.06
SPPP	Social Comparison	Stress	.20	.01	.02	-.02	.05

Note. CI = Confidence Interval; SOPP = Self-Orientated Parenting Perfectionism; SPPP = Societal-Prescribed Parenting Perfectionism.

and DASS-21 scores ($p > .05$, see Table 3). Additionally, the analyses revealed that Instagram frequency did not mediate the relationship between mothers with both dimensions of perfectionism (SOPP and SPPP) and DASS-21 scores ($p > .05$). However, contrary to prediction, there was a significant indirect effect between SPPP and depression *ab* = .02, BCa 95% CI [.00, .04], R^2 = .29 (moderate effect) through Facebook frequency, and between SPPP and anxiety *ab* = .02, BCa 95% CI [.00, .05] through Facebook frequency. The model accounted for 15% of the variance (R^2 = .15, moderate effect).

Hypothesis 4 predicted that the mental health of mothers with high standards placed on themselves to be a perfect parent (SOPP) would be mediated by comparing themselves with other mothers and families on social media, while controlling for social support. As can be seen in Table 4, Social Comparison was found to significantly mediate the relationship between mothers with SOPP and depression when controlling for social support (*ab* = .03, BCa 95% CI [.00, .07]). This model accounted for 25% (R^2 = .25, moderate effect) of the total variance. Additionally, the analyses showed that there was a significant indirect effect of mothers with SOPP and anxiety through social comparison, *ab* = .02, BCa 95% CI [.00, .05], R^2= .15 (moderate effect). However, contrary to prediction, social comparison did not mediate the relationship between mothers with SOPP and stress, *ab* = .02, BCa 95% CI [–.01, .05].

The hypotheses also predicted that the mental health of mothers with the belief that society expects them to be a perfect parent (SPPP) would be mediated by comparing themselves with other mothers and families on social media, while controlling for social support. As hypothesized, there was a significant indirect effect of mothers with SPPP and anxiety through social comparison (*ab* = .03, BCa 95% CI [.00, .06]), which accounted for 15% of the total effect (R^2 = .15), was evident. Contrary to prediction however, social comparison did not mediate the relationship between SPPP and depression and stress.

DISCUSSION

The present study aimed to examine whether perfectionism affects the mental health of mothers through the social comparison processes when using Facebook and Instagram. Given that social support is suggested to be a protective factor against the development of depression, anxiety, and stress in mothers (Haslam et al., 2006), the current study controlled for this.

The first hypothesis was supported, whereby both dimensions of perfectionism (SOPP and SPPP) in mothers were associated with depressive, anxious, and stress symptoms. This finding is consistent with existing research, which suggests that mothers with perfectionistic traits are more likely to display poorer parental adjustment, increased anxiety, and lowered self-efficacy (Gelabert et al., 2012; Lee et al., 2012).

Consistent with our second hypothesis the relationships between dimensions of perfectionism and maternal mental health remained significant but were weaker after controlling for the influence of social support. In other words, social support indeed has some protective value so far as depression, anxiety, and stress are concerned. However, the relationship between perfectionism and mental health is also independent of social support, suggesting that social support alone may not eliminate the caustic relationships between perfectionism and mental health in mothers.

The third hypothesis predicted that the frequency of social media use would not influence the relationship between moth-

ers with perfectionism and their mental health, after partialling out the influence of social support. As predicted, no relationship was found between Facebook frequency and the mental health of mothers with SOPP. Thus, the amount of time spent on Facebook did not appear to impact on the mental health of mothers with self-oriented parenting perfectionism. This is consistent with the study by Jelenchick et al. (2013), which found no relationship between mental health symptoms, specifically depression and the frequency of Facebook use. Additionally, as predicted, Facebook frequency did not influence symptoms of stress in mothers with SPPP, after controlling for social support. Contrary to prediction, however, Facebook frequency was found to influence depression and anxiety symptoms in mothers with SPPP. Like Schoppe-Sullivan et al. (2017), it is possible in this study that mothers with SPPP may increase their Facebook activity to gain external validation and/or support, and this negatively impacts on their mental health, especially when they may not receive validation for their parenting. Another possibility is that the increased exposure to profiles of others who promote the most positive representations of their lives reinforces the dominant discourse within society of the ideal mother (Dorethy et al., 2014; Henderson et al., 2016; Pempek et al., 2009). This then may lead to an increase in anxiety and depressive symptoms among mothers with SPPP who worry about how society may view their parenting compared to others.

As predicted, Instagram frequency was not associated with the mental health symptoms of mothers with either SOPP or SPPP. Given that Facebook is distinct from Instagram, Facebook may have a greater impact on mothers with SPPP as it combines both images and written text, rather than the sharing of images alone. Alternatively, Instagram is a newer social networking site and is not utilized by as many people as Facebook (Cowling, 2017). This was evident in the current study, whereby the average amount of time the sample spent on Facebook equalled one hour per day, compared to 17.9 minutes per day on Instagram.

Hypothesis 4 examined the role of comparing one's mothering with other mothers and families on social media, and the influence this may have on the mental health of mothers with SOPP

and SPPP, after controlling for social support. Indeed, after controlling for social support, social comparison was found to significantly mediate the relationship between mothers who placed high standards on themselves to be a perfect parent (SOPP), and their reported depression and anxiety symptoms. This suggests that mothers who already hold their own excessively high standards for parenting may use the motherhood ideal portrayed on Facebook and Instagram as a benchmark for comparison (Choi et al., 2005; Chou & Edge, 2012), resulting in symptoms of depression and anxiety. Thus, for this group of mothers, it would seem that it may not be the frequency of social media use which is harmful, but rather, the social comparative aspects of the platform which contributes to mental health problems.

These findings are in contrast to those of Lee et al. (2012) who found SOPP to be adaptive in mothers. However, Lee et al. (2012) measured perfectionism during the participants' third trimester of pregnancy and did not assess the influence of social media. Therefore, it could be argued that our findings may reflect different standards of perfectionism which might arise following pregnancy. Perhaps, after becoming a new parent, mothers are more likely to use Facebook and Instagram to compare their mothering to others. Given that "mothers are held responsible for not only their own wellbeing, but for their children's as well" (p. 515, Henderson et al., 2016), the process of social comparison with other mothers on social media may contribute to the belief that "I am a bad mother because I don't manage everything as well as they do" (p. 25, Djafarova & Trofimenko, 2017) and, in turn, symptoms of depression and anxiety.

Contrary to prediction however, the current study did not find an association with stress symptoms in mothers with SOPP who engaged in social comparison. This finding may be reflective of the questionable construct validity of the stress scale of the DASS-21. For instance, Antony, Bieling, Cox, Enns, and Swinson (1998), suggested that the stress scale was developed from items related to depression and anxiety rather than stress per se (Antony et al., 1998; Henry & Crawford, 2005).

There were mixed findings regarding the relationship between social comparison and mothers with societal-prescribed parent-

ing perfectionism (SPPP). As predicted, after controlling for social support, mothers with SPPP engaged in social comparison which contributed to their anxiety symptoms. However, social comparison did not mediate the relationship between mothers with SPPP and their reported depression and stress symptoms.

Since SPPP reflects a strong preoccupation with social judgement (Henderson et al., 2016), it may be that they did not disclose their dissatisfaction or distress associated motherhood when completing the questionnaire, for fear of being perceived as a "bad mother" (p. 176) by the researchers. Nevertheless, it appears that mothers who feel pressure to meet "society's impossible high standards for parenting" (p. 286, Schoppe-Sullivan et al., 2016) engage in social comparison while on social media, and this contributes to self-reported symptoms of anxiety.

The current study has several limitations which future research could address. First, the generalizability of our findings may be limited by our recruitment method (online convenience sampling) and the fact that the majority of the sample was Caucasian, more than three quarters of the sample had a university qualification and more than two thirds were employed. Second, given the sensitive nature of the topic and the transparency of the self-report items, participants may have responded in a socially desirable manner despite being informed of their anonymity. Third, the study included a sample of mothers of all ages, who had an unspecified number of children also of varying ages. Given that new mothers are the most active internet users (Plantin & Daneback, 2009) future studies may aim to determine whether the aforementioned relationship are especially strong in first-time mothers. Fourth, Facebook and Instagram frequency were measured using single-items, and the nature of social media engagement (e.g., active or passive) was not stipulated. Finally, the cross-sectional nature of our study precludes causal or temporal inferences about the relationships between our variables.

The present study is the first to investigate the extent to which Facebook and Instagram impact on the mental health of mothers with dimensions of perfectionism. Our findings provide insights regarding the potential risk factors that can impact the mental

health of mothers, noting that these can also be detrimental to the cognitive and social development of the child (Milgrom et al., 2004). Interventions aimed at educating mothers about unhelpful imagery presented on social media and the counterproductive effects of social comparison may help to mitigate this.

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